

Subash Sharma

Location: Bristol, UK

**Electron Microscopy Specialist | Materials
Scientist (2D materials, CVD, in situ TEM)**

University of Bristol, Bristol, UK

Google Scholar:
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ORCID: orcid.org/0000-0002-8783-3587

PROFILE

- Deep expertise in large-area 2D materials synthesis and process control (graphene, hBN, TMDCs) via chemical vapor deposition (CVD), including polymer-assisted transfer.
- Electron microscopy specialist with 12+ years of research experience across (S)TEM and in situ TEM (bias and heating), applied to advanced materials and nanocarbon systems.
- Google Scholar (September 2026): 1,077 citations; h-index 18; i10-index 29. Includes journal articles, conference contributions and other research outputs.
- Active peer reviewer for leading journals (e.g., Nano Letters, Nanoscale, Nanotechnology, Physica Status Solidi, Superlattices and Microstructures) with multiple reviewer awards (IOP/Elsevier).

CORE EXPERTISE

Synthesis & characterization

- CVD growth and transfer of graphene, hBN, MoS₂, WS₂ and related materials
- Process optimization and parameter control
- Thin-film/metal deposition tools (sputtering, ALD) and device-relevant processing
- Raman spectroscopy, AFM and SEM characterization

Electron microscopy

- Transmission electron microscopy (S/TEM) operation
- In situ TEM using bias voltage and heating holders
- In situ growth/assembly of nanostructures (graphene/CNT) and dynamic mechanism studies

PROFESSIONAL EXPERIENCE

Electron Microscopy Specialist | School of Chemistry, University of Bristol, Bristol, UK 2024/07

- Present

- Support advanced (S)TEM research projects and microscopy users; acquire and interpret high-quality datasets for materials/chemistry applications.
- Design and perform in situ TEM experiments (bias and heating) to observe growth, transformation and property evolution at the nanoscale.

- Train and advise students/researchers in microscopy workflows, data quality and experiment planning.

Advanced Furnace Technology, Cambridge, UK | R&D Engineer / Materials Process Engineer (Tantalum Carbide)

2023 - 2024

- Worked on large-scale growth and process development of tantalum carbide (TaC).
- Contributed to the design and build specification of a high-temperature furnace for scale-up (requirements, layout, process constraints, documentation).
- Bridged R&D and engineering: repeatability, throughput, safety, and practical manufacturability considerations.

Postdoctoral Fellow | Cambridge Graphene Centre, University of Cambridge, Cambridge, UK

2019 - 2023

- Led work packages within the Graphene Flagship programme, coordinating scalable 2D materials development through experimental planning, data delivery, milestone reporting, and collaboration with multidisciplinary academic and industrial partners.
- Developed and optimized wafer-scale CVD synthesis and transfer of graphene, hBN, and TMDCs, focusing on uniformity, repeatability, defect control, and device-ready integration.
- Collaborated across multidisciplinary teams; contributed to peer-reviewed publications and dissemination.
- Supervised and guided PhD students at the Cambridge Graphene Centre (experimental planning, TEM training, data analysis, and publication support).

Postdoctoral Fellow | Department of Physical Science and Engineering, Nagoya Institute of Technology, Nagoya, Japan

April 2015 - November 2019

- Developed CVD synthesis and transfer processes for graphene and hBN; investigated growth mechanisms and morphology control.
- Performed in situ TEM synthesis/assembly of graphene and CNT structures; studied nanoscale transformation processes.

Researcher (PhD Research) | Department of Physical Science and Engineering, Nagoya Institute of Technology, Nagoya, Japan

2012 - March 2015 (Advisor: Prof. Masaki Tanemura)

- PhD research focused on synthesis of graphene and hexagonal boron nitride (hBN) using solid precursors by chemical vapor deposition (CVD).

Graduate Student | Department of Electrical and Electronic Engineering, Chubu University, Kasugai, Japan 2010 - 2012 (Advisor: Prof. Shunjiro Ikezawa)

- Deposition of TiO₂ films using plasma processes; investigated thin-film properties (TiN/TiO₂ systems).

EDUCATION

PhD in Engineering | Nagoya Institute of Technology, Nagoya, Japan

March 2015

- Dissertation: "Synthesis of graphene and hexagonal boron nitride (hBN) using solid precursors by chemical vapor deposition (CVD)"

Master of Engineering | Chubu University, Kasugai, Japan

March 2012

- Dissertation: "Production of TiN/TiO₂ films and its properties"

BSc in Physics | Tribhuvan University, Chitwan, Nepal

2008

TEACHING & MENTORING

Teaching Assistant | Nagoya Institute of Technology, Nagoya, Japan

2012 - 2015

- Course: Introduction of Solid State Physics.

Teacher (Mathematics & Science) | Hillbird Higher Secondary School, Chitwan, Nepal 2007

- 2008

Research Mentor | Nagoya Institute of Technology, Nagoya, Japan 2015

- 2019

- Mentored undergraduate and graduate students on research projects and experimental methods.

AWARDS & RECOGNITION

- Outstanding research achievement and contribution award (Green 2017); invited presentation, Taipei, Taiwan.
- Outstanding Reviewer Awards (2017), Nanotechnology (IOP).
- Certificate of Outstanding Contribution in Reviewing (2017), Superlattices and Microstructures (Elsevier).

PROFESSIONAL SERVICE

- Journal reviewer: Nano Letters; Nanoscale; Applied Materials and Interfaces; Nanotechnology; Physica Status Solidi; Superlattices and Microstructures; and related journals.

SELECTED PUBLICATIONS (HIGH IMPACT)

1. S. Sharma, G. Kalita, R. Hirano, S.M. Shinde, R. Papon, H. Ohtani, et al., "Synthesis of graphene crystals from solid waste plastic by chemical vapor deposition," Carbon 72, 66-73 (2014).
2. G. Kalita, M.E. Ayhan, S. Sharma, S.M. Shinde, D. Ghimire, K. Wakita, et al., "Low temperature deposited graphene by surface wave plasma CVD as effective oxidation resistive barrier," Corrosion Science 78, 183-187 (2014).
3. S. Sharma, G. Kalita, R. Vishwakarma, Z. Zulkifli, M. Tanemura, "Opening of triangular hole in triangularshaped CVD hBN crystal," Scientific Reports 5, 10426 (2015).

4. S. Sharma, G. Kalita, R. Hirano, Y. Hayashi, M. Tanemura, "Influence of gas composition on the formation of graphene domain synthesized from camphor," *Materials Letters* 93, 258-262 (2013).
5. G. Kalita, S. Sharma, K. Wakita, M. Umeno, Y. Hayashi, M. Tanemura, "A photoinduced charge transfer composite of graphene oxide and ferrocene," *Physical Chemistry Chemical Physics* 15, 1271-1274 (2013).
6. K.P. Sharma, S.M. Shinde, M.S. Rosmi, S. Sharma, G. Kalita, M. Tanemura, "Effect of copper foil annealing process on large graphene domain growth by solid source-based CVD," *Journal of Materials Science* 51, 7220-7228 (2016).
7. M.S. Rosmi, S.M. Shinde, N.D.A. Rahman, A. Thangaraja, S. Sharma, et al., "Monolayer graphene on resolidified Cu from waste chicken fat by LP-CVD," *Materials Research Bulletin* 83, 573-580 (2016).
8. S. Sharma, M.S. Rosmi, Y. Yaakob, M.Z.M. Yusop, G. Kalita, et al., "In situ TEM synthesis of Y-junction carbon nanotube by electromigration induced soldering," *Carbon* 132, 165-171 (2018).
9. K.P. Sharma, S. Sharma, A.K. Sharma, B.P. Jaisi, G. Kalita, M. Tanemura, "Edge controlled growth of hBN crystals on copper foil by atmospheric pressure CVD," *CrystEngComm* 20, 550-555 (2018).
10. M. Asgari, L. Viti, O. Balci, S.M. Shinde, J. Zhang, H. Ramezani, S. Sharma, et al., "Terahertz photodetection in scalable single-layer-graphene and hBN heterostructures," *Applied Physics Letters* 121(3) (2022).
11. E. Calandrini, K. Voronin, O. Balci, M.B. Burillo, A. Bylinkin, S.M. Shinde, S. Sharma, et al., "Near- and FarField Observation of Phonon Polaritons in wafer-scale multilayer hBN prepared by CVD," *Advanced Materials* (2023).
12. C. Xu, K. Li, S. Liu, J. Xu, S. Sharma, et al., "Identification of Cu-N₂ Sites for Zn-Air Batteries in Harsh Electrolytes: Computer Virtual Screening, Machine Learning, and Practical Application," *CCS Chemistry* (2025).

PUBLICATIONS (FULL LIST)

- 1) S. Elnobi, S. Sharma, T. Osugi, B.P. Jaisi, G. Kalita, M.Z.M. Yusop, M.E. Ayhan, Z.Q.C. Ng, D.H.C. Chua, M. Tanemura, "One-step synthesis of spontaneously graphitized nanocarbon using cobalt-nanoparticles," *SN Applied Sciences* 2, 2147 (2020).
- 2) S. Sharma*, T. Osugi, S. Elnobi, S. Ozeki, B.P. Jaisi, G. Kalita, C. Capiglia, M. Tanemura, "Synthesis and Characterization of Li-C Nanocomposite for Easy and Safe Handling," *Nanomaterials* 10, 1483 (2020).
- 3) B.P. Jaisi, S. Sharma, S. Elnobi, A.A. Abuelwafa, Y. Yaakob, G. Kalita, M. Tanemura, "Graphitization of GalliumIncorporated Carbon Nanofibers and Cones: In Situ and Ex Situ TEM Studies," *Physica Status Solidi (b)* 257, 2000309 (2020).
- 4) S. Elnobi, S. Sharma, M.I. Araby, B. Paudel, G. Kalita, M.Z.M. Yusop, M.E. Ayhan, M. Tanemura, "Room-temperature graphitization in a solid-phase reaction," *RSC Advances* 10, 914-922 (2020).
- 5) M.I. Araby, S. Sharma, S. Elnobi, G. Kalita, M. Kitazawa, M. Tanemura, "Temperature dependence of catalytic activity in graphene synthesis for Sn nanoparticles," *Journal of Materials Science: Materials in Electronics* 30, 12796-12803 (2019).

- 6) S. Sharma*, B.P. Jaisi, M.I. Araby, S. Elnobi, M.E. Ayhan, G. Kalita, M. Tanemura, "The Mo catalyzed graphitization of amorphous carbon: an in situ TEM study," *RSC Advances* 9, 34377-34381 (2019).
- 7) K. Watanabe, S. Sharma, M.I. Araby, M. Shima, T. Harada, M.S. Rosmi, G. Kalita, M. Kitazawa, M. Tanemura, "Development of oxide nanofiber-tipped cantilever as a substrate for cross-sectional TEM analysis," *Surface and Interface Analysis* 50, 1122-1126 (2018).
- 8) B.P. Jaisi, K.P. Sharma, S. Sharma, R.D. Mahayavanshi, G. Kalita, M. Tanemura, "Switching isotropic and anisotropic graphene growth in a solid source CVD system," *CrystEngComm* 20, 5356-5363 (2018).
- 9) S. Sharma*, M.S. Rosmi, Y. Yaakob, M.Z.M. Yusop, G. Kalita, M. Kitazawa, M. Tanemura, "In situ TEM synthesis of Y-junction carbon nanotube by electromigration induced soldering," *Carbon* 132, 165-171 (2018).
- 10) K.P. Sharma, S. Sharma, A.K. Sharma, B.P. Jaisi, G. Kalita, M. Tanemura, "Edge controlled growth of hexagonal boron nitride crystals on copper foil by atmospheric pressure chemical vapor deposition," *CrystEngComm* 20, 550-555 (2018).
- 11) S. Sharma*, B.P. Jaisi, K.P. Sharma, M.I. Araby, G. Kalita, M. Tanemura, "Synthesis of Freestanding WS₂ Trees and Fibers on Au by Chemical Vapor Deposition (CVD)," *Physica Status Solidi (a)* 215, 1700566 (2018).
- 12) M.I. Araby, M.S. Rosmi, R. Vishwakarma, S. Sharma, Y. Wakamatsu, K. Takahashi, G. Kalita, M. Kitazawa, M. Tanemura, "Graphene formation at 150 C using indium as catalyst," *RSC Advances* 7, 47353-47356 (2017).
- 13) R. Papon, C. Pierlot, S. Sharma, S.M. Shinde, G. Kalita, M. Tanemura, "Optimization of CVD parameters for graphene synthesis through design of experiments," *Physica Status Solidi (b)* 254, 1600629 (2017).
- 14) R. Papon, K.P. Sharma, R.D. Mahayavanshi, S. Sharma, R. Vishwakarma, M.S. Rosmi, T. Kawahara, J. Cline, G. Kalita, M. Tanemura, "CuNi binary alloy catalyst for growth of nitrogen-doped graphene by low pressure chemical vapor deposition," *Physica Status Solidi (RRL)* 10, 749-752 (2016).
- 15) S. Sharma*, K. Sharma, M.S. Rosmi, Y. Yaakob, M.I. Araby, H. Ohtani, G. Kalita, M. Tanemura, "Morphology-controlled synthesis of hexagonal boron nitride crystals by chemical vapor deposition," *Crystal Growth & Design* 16, 6440-6445 (2016).
- 16) M.S. Rosmi, Y. Yaakob, M.Z. Mohd Yusop, S. Sharma, Z. Zulkifli, A. Supee, G. Kalita, M. Tanemura, "Room temperature fabrication of 1D carbon-copper composite nanostructures directly on Cu substrate and their field emission properties," *AIP Advances* 6, 095109 (2016).
- 17) M.S. Rosmi, Y. Yaakob, M.Z.M. Yusop, S. Sharma, R. Vishwakarma, M.I. Araby, G. Kalita, M. Tanemura, "In situ fabrication of graphene from a copper-carbon nanoneedle and its electrical properties," *RSC Advances* 6, 82459-82466 (2016).
- 18) M.S. Rosmi, S.M. Shinde, N.D.A. Rahman, A. Thangaraja, S. Sharma, K.P. Sharma, Y. Yaakob, R.K. Vishwakarma, S.A. Bakar, G. Kalita, "Synthesis of uniform monolayer graphene on re-solidified copper from waste chicken fat by low pressure chemical vapor deposition," *Materials Research Bulletin* 83, 573-580 (2016).
- 19) M.S. Rosmi, Y. Yaakob, S. Sharma, M.Z. Yusop, G. Kalita, M. Tanemura, "In situ TEM visualization of Pd assisted graphene growth in nanoscale," *IEEE 16th International Conference on Nanotechnology (IEEE-NANO)*, 622-623 (2016).
- 20) K.P. Sharma, S.M. Shinde, M.S. Rosmi, S. Sharma, G. Kalita, M. Tanemura, "Effect of copper foil annealing process on large graphene domain growth by solid source-based chemical vapor deposition," *Journal of Materials Science* 51, 7220-7228 (2016).
- 21) R. Vishwakarma, S. Sharma, S.M. Shinde, K.P. Sharma, A. Thangaraja, G. Kalita, M. Tanemura, "Fabrication of particular structures of hexagonal boron nitride and boron-carbon-nitrogen layers by anisotropic etching," *Physica E: Low-dimensional Systems and Nanostructures* 79, 13-19 (2016).
- 22) Z. Zulkifli, S. Sharma, S. Shinde, G. Kalita, M. Tanemura, "Effect of annealing in hydrogen atmosphere on ZnO films for field emission display," *IOP Conference Series: Materials Science and Engineering* 99, 012030 (2015).
- 23) A. Thangaraja, S.M. Shinde, G. Kalita, R. Papon, S. Sharma, R. Vishwakarma, K.P. Sharma, M. Tanemura, "Structure dependent hydrogen induced etching features of graphene crystals," *Applied Physics Letters* 106, 253106 (2015).
- 24) S. Sharma*, G. Kalita, R. Vishwakarma, Z. Zulkifli, M. Tanemura, "Opening of triangular hole in triangular-shaped chemical vapor deposited hexagonal boron nitride crystal," *Scientific Reports* 5, 10426 (2015).
- 25) S.M. Shinde, G. Kalita, S. Sharma, Z. Zulkifli, R. Papon, M. Tanemura, "Polymer-free graphene transfer on moldable cellulose acetate based paper by hot press technique," *Surface and Coatings Technology* 275, 369-373 (2015).
- 26) R. Papon, S. Sharma, S.M. Shinde, A. Thangaraja, G. Kalita, M. Tanemura, "Formation of graphene nanoribbons and Y-junctions by hydrogen induced anisotropic etching," *RSC Advances* 5, 35297-35301 (2015).

- 27) R. Papon, G. Kalita, S. Sharma, S.M. Shinde, R. Vishwakarma, M. Tanemura, "Controlling single and few-layer graphene crystals growth in a solid carbon source based chemical vapor deposition," *Applied Physics Letters* 105, 133103 (2014).
- 28) G. Kalita, R. Papon, S. Sharma, S.M. Shinde, R. Vishwakarma, M. Tanemura, "Transformation of chemical vapor deposited individual graphene crystal with oxidation of copper substrate," *Carbon* 80, 504-512 (2014).
- 29) S. Sharma, G. Kalita, R. Hirano, S.M. Shinde, R. Papon, H. Ohtani, M. Tanemura, "Synthesis of graphene crystals from solid waste plastic by chemical vapor deposition," *Carbon* 72, 66-73 (2014).
- 30) S.M. Shinde, G. Kalita, S. Sharma, R. Papon, M.Z. Yusop, M. Tanemura, "Synthesis of a three dimensional structure of vertically aligned carbon nanotubes and graphene from a single solid carbon source," *RSC Advances* 4, 13355-13360 (2014).
- 31) M.E. Ayhan, G. Kalita, S. Sharma, M. Tanemura, "Chemical vapor deposition of graphene on silver foil as a tarnishresistant coating," *Physica Status Solidi (RRL)* 7, 1076-1079 (2013).
- 32) G. Kalita, M.E. Ayhan, S. Sharma, S.M. Shinde, D. Ghimire, K. Wakita, M. Umeno, M. Tanemura, "Low temperature deposited graphene by surface wave plasma CVD as effective oxidation resistive barrier," *Corrosion Science* 78, 183187 (2014).
- 33) S. Sharma, G. Kalita, "Synthesis of hexagonal graphene on polycrystalline Cu foil from solid camphor by atmospheric pressure chemical vapor deposition," *Journal of Materials Science* 48, 7036-7041 (2013).
- 34) S. Sharma, G. Kalita, R. Hirano, Y. Hayashi, M. Tanemura, "Influence of gas composition on the formation of graphene domain synthesized from camphor," *Materials Letters* 93, 258-262 (2013).
- 35) G. Kalita, S. Sharma, K. Wakita, M. Umeno, Y. Hayashi, M. Tanemura, "A photoinduced charge transfer composite of graphene oxide and ferrocene," *Physical Chemistry Chemical Physics* 15, 1271-1274 (2013).
- 36) G. Kalita, S. Sharma, K. Wakita, M. Umeno, Y. Hayashi, M. Tanemura, "Synthesis of graphene by surface wave plasma chemical vapor deposition from camphor," *Physica Status Solidi (a)* 209, 2510-2513 (2012).
- 37) M. Asgari, L. Viti, O. Balci, S.M. Shinde, J. Zhang, H. Ramezani, S. Sharma, et al., "Terahertz photodetection in scalable single-layer-graphene and hexagonal boron nitride heterostructures," *Applied Physics Letters* 121(3) (2022).
- 38) E. Calandrini, K. Voronin, O. Balci, M.B. Burillo, A. Bylinkin, S.M. Shinde, S. Sharma, et al., "Near- and Far-Field Observation of Phonon Polaritons in wafer scale multilayer hexagonal boron nitride prepared by chemical vapor deposition," *Advanced Materials*, 2302045 (2023).
- 39) C. Xu, K. Li, S. Liu, J. Xu, S. Sharma, et al., "Identification of Cu-N₂ Sites for Zn-Air Batteries in Harsh Electrolytes: Computer Virtual Screening, Machine Learning, and Practical Application," *CCS Chemistry* (2025).
- 40) M. Tiberi, A. Montanaro, C. Wen, J. Zhang, O. Balci, S.M. Shinde, S. Sharma, et al., "Graphene Electro-Absorption Modulators for Energy-Efficient and High-Speed Optical Transceivers," *arXiv:2506.03281* (2025).
- 41) S. Giri, S. Sharma, R.D. Mahayavanshi, G. Kalita, Y. Yang, M. Tanemura, "In Situ TEM Study of Electrical Property and Mechanical Deformation in MoS₂/Graphene Heterostructures," *Nanomaterials* 15(2), 114 (2025).

(* indicates corresponding author, as marked in the original CV list where applicable.)